



Jurusan Teknologi Informasi Politeknik Negeri Malang
Mata Kuliah Data Warehouse
Kuis 1

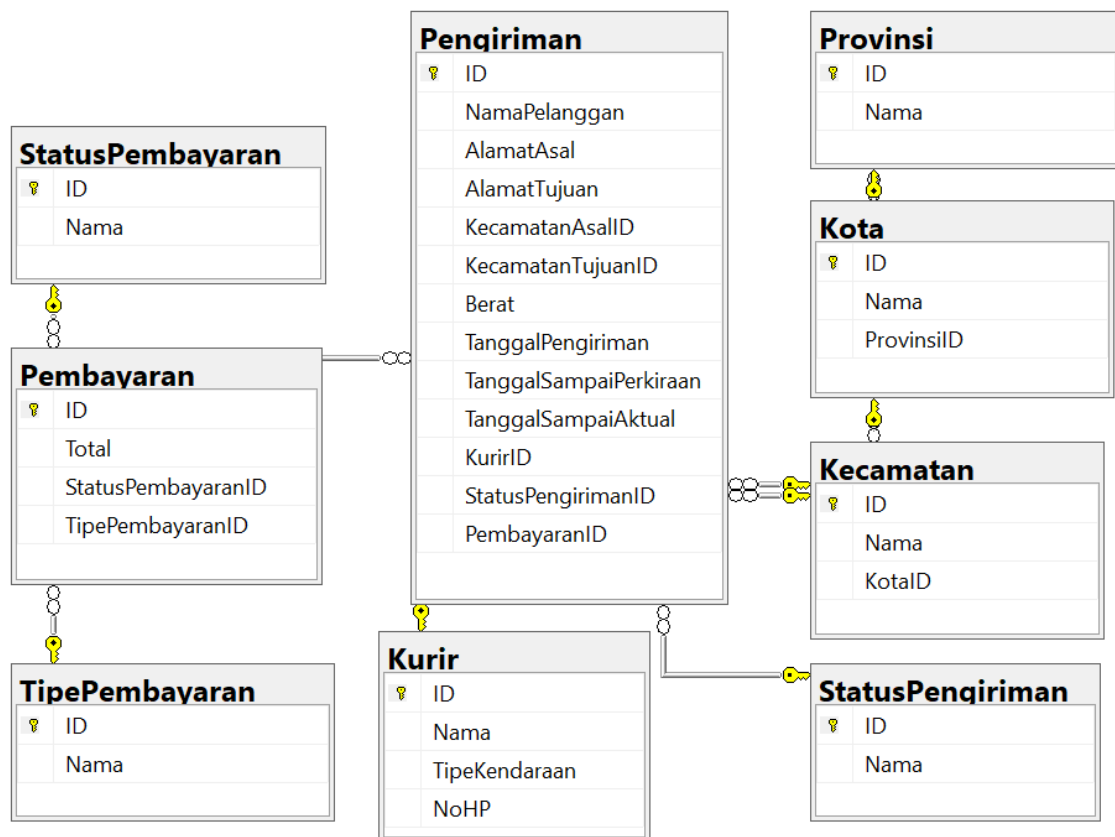
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Nomor Urut : 09

1. Tuliskan perbandingan star schema dan snowflake schema pada tabel berikut:

	Star Schema	Snowflake Schema
Normalisasi	Denormalisasi (tabel fakta dan dimensi dirancang dengan sedikit normalisasi).	Normalisasi (tabel dimensi dipecah menjadi beberapa tabel terkait).
Kompleksitas desain/skema	Sederhana (struktur hierarki datar dengan satu tabel fakta dan beberapa dimensi).	Lebih kompleks (tabel dimensi dipecah menjadi beberapa tabel yang saling terkait).
Kompleksitas query	Lebih sederhana (query melibatkan sedikit join karena denormalisasi).	Lebih kompleks (query memerlukan lebih banyak join karena normalisasi).
Performa query	Lebih cepat (karena denormalisasi dan sedikit join).	Lebih lambat (karena normalisasi dan lebih banyak join).
Storage	Lebih besar (karena duplikasi data pada tabel dimensi).	Lebih kecil (karena normalisasi mengurangi duplikasi data).
Integritas data	Lebih rendah (karena denormalisasi, risiko inkonsistensi data lebih tinggi).	Lebih tinggi (karena normalisasi, data lebih konsisten).
Maintenance (pengisian data dengan proses ETL dari OLTP)	Lebih mudah (struktur sederhana, proses ETL lebih cepat).	Lebih sulit (struktur kompleks, proses ETL memerlukan lebih banyak transformasi).

2. Gambar berikut menunjukkan skema OLTP database dari sebuah sistem informasi ekspedisi. Buatlah data warehouse dalam star schema yang digunakan sebagai dasar analisis performa ekspedisi.



1. Membuat database

- Create Table
 - tr_Pengiriman

```
CREATE TABLE tr_Pengiriman (  
  Kode_Pengiriman INT PRIMARY KEY,  
  NamaPelanggan VARCHAR(255),  
  AlamatAsal VARCHAR(255),  
  AlamatTujuan VARCHAR(255),  
  Kode_kecamatanAsal INT,  
  Kode_kecamatanTujuan INT,  
  Berat DECIMAL(10, 2),  
  TanggalPengiriman DATE,  
  TanggalSampaiPerkiraan DATE,  
  TanggalSampaiAktual DATE,  
  Kode_kurir INT,  
  Kode_statuspengiriman INT,  
  Kode_pembayaran INT,  
  FOREIGN KEY (Kode_kecamatanAsal) REFERENCES ms_Kecamatan(Kode_kecamatan),  
  FOREIGN KEY (Kode_kecamatanTujuan) REFERENCES ms_Kecamatan(Kode_kecamatan),  
  FOREIGN KEY (Kode_kurir) REFERENCES ms_Kurir(Kode_kurir),  
  FOREIGN KEY (Kode_statuspengiriman) REFERENCES  
ms_StatusPengiriman(Kode_statuspengiriman),  
  FOREIGN KEY (Kode_pembayaran) REFERENCES ms_Pembayaran(Kode_pembayaran)  
);
```

```
CREATE TABLE ms_StatusPembayaran (  
  Kode_statuspembayaran INT PRIMARY KEY,  
  Nama VARCHAR(255)  
);
```

- ms_Pembayaran

```
CREATE TABLE ms_Pembayaran (  
  Kode_pembayaran INT PRIMARY KEY,  
  Total DECIMAL(10, 2),  
  Kode_statuspembayaran INT,  
  Kode_tipepembayaran INT,  
  FOREIGN KEY (Kode_statuspembayaran) REFERENCES  
ms_StatusPembayaran(Kode_statuspembayaran),  
  FOREIGN KEY (Kode_tipepembayaran) REFERENCES  
ms_TipePembayaran(Kode_tipepembayaran)  
);
```

- ms_TipePembayaran

```
CREATE TABLE ms_TipePembayaran (  
  Kode_tipepembayaran INT PRIMARY KEY,  
  Nama VARCHAR(255)  
);
```

- ms_Kurir

```
CREATE TABLE ms_Kurir (  
  Kode_kurir INT PRIMARY KEY,  
  Nama VARCHAR(255),  
  TipeKendaraan VARCHAR(255),  
  NoHP VARCHAR(15)  
);
```

- ms_StatusPengiriman

```
CREATE TABLE ms_StatusPengiriman (  
  Kode_statuspengiriman INT PRIMARY KEY,  
  Nama VARCHAR(255)  
);
```

- ms_Kecamatan

```
CREATE TABLE ms_Kecamatan (  
  Kode_kecamatan INT PRIMARY KEY,  
  Nama VARCHAR(255),  
  Kode_kota INT,  
  FOREIGN KEY (Kode_kota) REFERENCES ms_Kota(Kode_kota)  
);
```

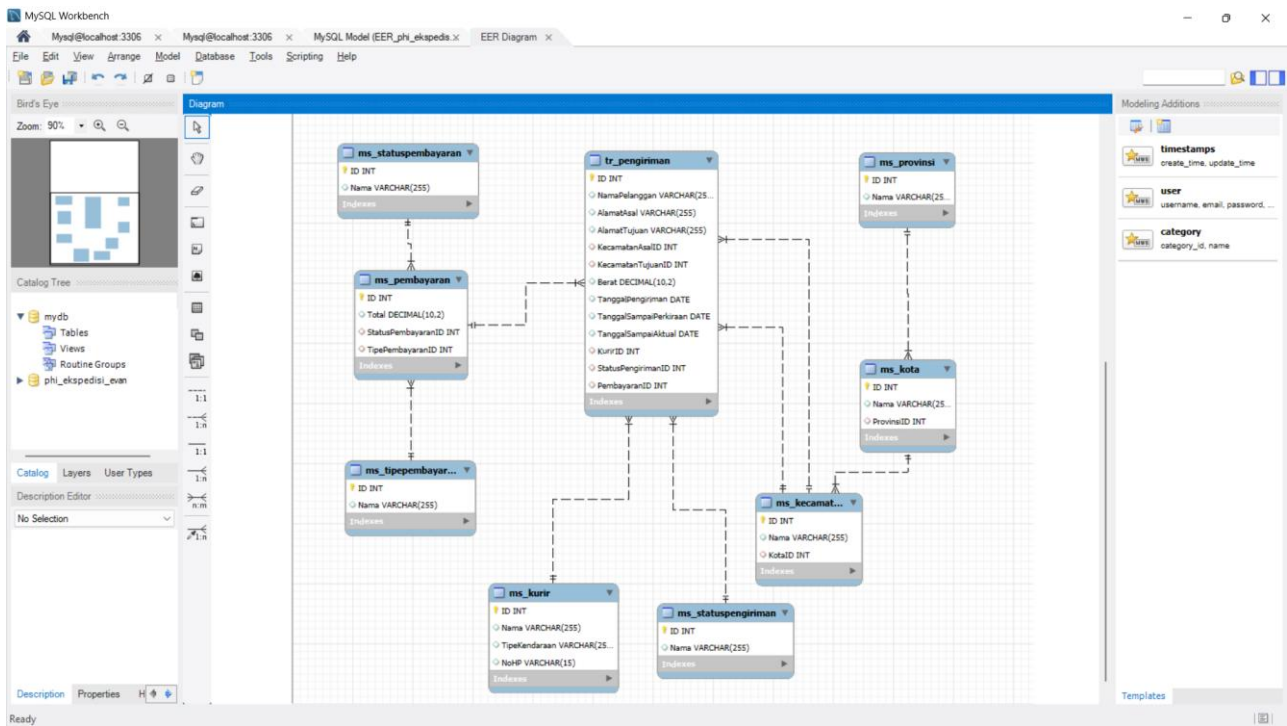
- ms_Kota

```
CREATE TABLE ms_Kota (  
  Kode_kota INT PRIMARY KEY,  
  Nama VARCHAR(255),  
  Kode_provinsi INT,  
  FOREIGN KEY (Kode_provinsi) REFERENCES ms_Provinsi(Kode_provinsi)  
);
```

- ms_Provinsi

```
CREATE TABLE ms_Provinsi (  
  Kode_provinsi INT PRIMARY KEY,  
  Nama VARCHAR(255)  
);
```

- Hasil Create Table



phi_ekspedisi_evan

- Insert Data
 - tr_Pengiriman

```

INSERT INTO tr_Pengiriman (Kode_pengiriman, NamaPelanggan, AlamatAsal, AlamatTujuan,
Kode_kecamatanAsal, Kode_kecamatanTujuan, Berat, TanggalPengiriman,
TanggalSampaiPerkiraan, TanggalSampaiAktual, Kode_kurir, Kode_statuspengiriman,
Kode_pembayaran) VALUES
(1, 'Pelanggan 1', 'Jl. Merdeka No. 1', 'Jl. Sudirman No. 2', 1, 2, 5.0, '2023-10-01', '2023-10-03',
'2023-10-03', 1, 1, 1),
(2, 'Pelanggan 2', 'Jl. Pahlawan No. 3', 'Jl. Gatot Subroto No. 4', 3, 4, 3.5, '2023-10-02', '2023-
10-04', '2023-10-04', 2, 3, 2),
(3, 'Pelanggan 3', 'Jl. Diponegoro No. 5', 'Jl. Ahmad Yani No. 6', 5, 6, 4.0, '2023-10-03', '2023-10-
05', '2023-10-05', 3, 2, 3),
(4, 'Pelanggan 4', 'Jl. Gajah Mada No. 7', 'Jl. Hayam Wuruk No. 8', 7, 8, 2.0, '2023-10-04', '2023-
10-06', '2023-10-06', 4, 1, 4),
(5, 'Pelanggan 5', 'Jl. Thamrin No. 9', 'Jl. Kebon Sirih No. 10', 9, 1, 6.0, '2023-10-05', '2023-10-
07', '2023-10-07', 5, 3, 5),
(6, 'Pelanggan 6', 'Jl. Sudirman No. 11', 'Jl. MH Thamrin No. 12', 2, 3, 3.0, '2023-10-06', '2023-
10-08', '2023-10-08', 6, 2, 6),
(7, 'Pelanggan 7', 'Jl. Gatot Subroto No. 13', 'Jl. Pahlawan No. 14', 4, 5, 4.5, '2023-10-07', '2023-
10-09', '2023-10-09', 7, 1, 7),
(8, 'Pelanggan 8', 'Jl. Ahmad Yani No. 15', 'Jl. Diponegoro No. 16', 6, 7, 5.5, '2023-10-08', '2023-
10-10', '2023-10-10', 8, 3, 8),
(9, 'Pelanggan 9', 'Jl. Hayam Wuruk No. 17', 'Jl. Gajah Mada No. 18', 8, 9, 2.5, '2023-10-09',
'2023-10-11', '2023-10-11', 9, 2, 9),
(10, 'Pelanggan 10', 'Jl. Kebon Sirih No. 19', 'Jl. Thamrin No. 20', 1, 2, 3.0, '2023-10-10', '2023-
10-12', '2023-10-12', 10, 1, 10),
(11, 'Pelanggan 11', 'Jl. Merdeka No. 21', 'Jl. Sudirman No. 22', 2, 3, 4.0, '2023-10-11', '2023-10-
13', '2023-10-13', 1, 3, 11),
(12, 'Pelanggan 12', 'Jl. Pahlawan No. 23', 'Jl. Gatot Subroto No. 24', 3, 4, 5.0, '2023-10-12',
'2023-10-14', '2023-10-14', 2, 2, 12),
(13, 'Pelanggan 13', 'Jl. Diponegoro No. 25', 'Jl. Ahmad Yani No. 26', 4, 5, 6.0, '2023-10-13',
'2023-10-15', '2023-10-15', 3, 1, 13),
(14, 'Pelanggan 14', 'Jl. Gajah Mada No. 27', 'Jl. Hayam Wuruk No. 28', 5, 6, 3.5, '2023-10-14',
'2023-10-16', '2023-10-16', 4, 3, 14),
(15, 'Pelanggan 15', 'Jl. Thamrin No. 29', 'Jl. Kebon Sirih No. 30', 6, 7, 4.5, '2023-10-15', '2023-
10-17', '2023-10-17', 5, 2, 15),
(16, 'Pelanggan 16', 'Jl. Sudirman No. 31', 'Jl. MH Thamrin No. 32', 7, 8, 2.0, '2023-10-16',
'2023-10-18', '2023-10-18', 6, 1, 16),
(17, 'Pelanggan 17', 'Jl. Gatot Subroto No. 33', 'Jl. Pahlawan No. 34', 8, 9, 3.0, '2023-10-17',
'2023-10-19', '2023-10-19', 7, 3, 17),
(18, 'Pelanggan 18', 'Jl. Ahmad Yani No. 35', 'Jl. Diponegoro No. 36', 9, 1, 4.0, '2023-10-18',
'2023-10-20', '2023-10-20', 8, 2, 18),
(19, 'Pelanggan 19', 'Jl. Hayam Wuruk No. 37', 'Jl. Gajah Mada No. 38', 1, 2, 5.0, '2023-10-19',
'2023-10-21', '2023-10-21', 9, 1, 19),
(20, 'Pelanggan 20', 'Jl. Kebon Sirih No. 39', 'Jl. Thamrin No. 40', 2, 3, 6.0, '2023-10-20', '2023-
10-22', '2023-10-22', 10, 3, 20),
(21, 'Pelanggan 21', 'Jl. Merdeka No. 41', 'Jl. Sudirman No. 42', 3, 4, 3.5, '2023-10-21', '2023-10-
23', '2023-10-23', 1, 2, 21),
(22, 'Pelanggan 22', 'Jl. Pahlawan No. 43', 'Jl. Gatot Subroto No. 44', 4, 5, 4.0, '2023-10-22',
'2023-10-24', '2023-10-24', 2, 1, 22),
(23, 'Pelanggan 23', 'Jl. Diponegoro No. 45', 'Jl. Ahmad Yani No. 46', 5, 6, 5.0, '2023-10-23',
'2023-10-25', '2023-10-25', 3, 3, 23),

```

```
(24, 'Pelanggan 24', 'Jl. Gajah Mada No. 47', 'Jl. Hayam Wuruk No. 48', 6, 7, 6.0, '2023-10-24',
'2023-10-26', '2023-10-26', 4, 2, 24),
(25, 'Pelanggan 25', 'Jl. Thamrin No. 49', 'Jl. Kebon Sirih No. 50', 7, 8, 2.5, '2023-10-25', '2023-
10-27', '2023-10-27', 5, 1, 25),
(26, 'Pelanggan 26', 'Jl. Sudirman No. 51', 'Jl. MH Thamrin No. 52', 8, 9, 3.5, '2023-10-26',
'2023-10-28', '2023-10-28', 6, 3, 26),
(27, 'Pelanggan 27', 'Jl. Gatot Subroto No. 53', 'Jl. Pahlawan No. 54', 9, 1, 4.5, '2023-10-27',
'2023-10-29', '2023-10-29', 2, 3, 27),
(28, 'Pelanggan 28', 'Jl. Ahmad Yani No. 55', 'Jl. Diponegoro No. 56', 1, 2, 5.5, '2023-10-28',
'2023-10-30', '2023-10-30', 8, 1, 28),
(29, 'Pelanggan 29', 'Jl. Hayam Wuruk No. 57', 'Jl. Gajah Mada No. 58', 2, 3, 6.5, '2023-10-29',
'2023-10-31', '2023-10-31', 9, 3, 29),
(30, 'Pelanggan 30', 'Jl. Kebon Sirih No. 59', 'Jl. Thamrin No. 60', 3, 4, 3.0, '2023-10-30', '2023-
11-01', '2023-11-01', 10, 2, 30);
```

- ms_StatusPembayaran

```
INSERT INTO ms_StatusPembayaran (Kode_statuspembayaran, Nama) VALUES
(1, 'Lunas'),
(2, 'Belum Lunas');
```

- ms_Pembayaran

```
INSERT INTO ms_Pembayaran (Kode_pembayaran, Total, Kode_statuspembayaran,
Kode_tipepembayaran) VALUES
(1, 150000, 1, 1),
(2, 200000, 1, 2),
(3, 250000, 2, 3),
(4, 300000, 1, 1),
(5, 350000, 1, 2),
(6, 400000, 2, 3),
(7, 450000, 1, 1),
(8, 500000, 1, 2),
(9, 550000, 2, 3),
(10, 600000, 1, 1),
(11, 650000, 1, 2),
(12, 700000, 2, 3),
(13, 750000, 1, 1),
(14, 800000, 1, 2),
(15, 850000, 2, 3),
(16, 900000, 1, 1),
(17, 950000, 1, 2),
(18, 1000000, 2, 3),
(19, 1050000, 1, 1),
(20, 1100000, 1, 2),
(21, 1150000, 2, 3),
(22, 1200000, 1, 1),
(23, 1250000, 1, 2),
(24, 1300000, 2, 3),
(25, 1350000, 1, 1),
(26, 1400000, 1, 2),
(27, 1450000, 2, 3),
(28, 1500000, 1, 1),
(29, 1550000, 1, 2),
(30, 1600000, 2, 3);
```

- ms_TipePembayaran

```
INSERT INTO ms_TipePembayaran (Kode_tipepembayaran, Nama) VALUES
(1, 'Transfer Bank'),
(2, 'Kartu Kredit'),
(3, 'Tunai');
```

- ms_Kurir

```
INSERT INTO ms_Kurir (Kode_kurir, Nama, TipeKendaraan, NoHP) VALUES
(1, 'Andi', 'Motor', '081234567890'),
(2, 'Budi', 'Mobil', '081234567891'),
(3, 'Cici', 'Motor', '081234567892'),
(4, 'Dedi', 'Mobil', '081234567893'),
(5, 'Eka', 'Motor', '081234567894'),
(6, 'Fani', 'Mobil', '081234567895'),
(7, 'Gani', 'Motor', '081234567896'),
(8, 'Hani', 'Mobil', '081234567897'),
(9, 'Ika', 'Motor', '081234567898'),
(10, 'Joni', 'Mobil', '081234567899');
```

- ms_StatusPengiriman

```
INSERT INTO ms_StatusPengiriman (Kode_statuspengiriman, Nama) VALUES
(1, 'Dalam Proses'),
(2, 'Dalam Pengiriman'),
(3, 'Ter kirim');
```

- ms_Kecamatan

```
INSERT INTO ms_Kecamatan (Kode_kecamatan, Nama, Kode_kota) VALUES
(1, 'Coblong', 1),
(2, 'Sukajadi', 1),
(3, 'Harjamukti', 2),
(4, 'Kejaksan', 2),
(5, 'Semarang Tengah', 3),
(6, 'Banjarsari', 4),
(7, 'Jebres', 4),
(8, 'Gubeng', 5),
(9, 'Klojen', 6);
```

- ms_Kota

```
INSERT INTO ms_Kota (Kode_kota, Nama, Kode_provinsi) VALUES
(1, 'Bandung', 1),
(2, 'Cirebon', 1),
(3, 'Semarang', 2),
(4, 'Surakarta', 2),
(5, 'Surabaya', 3),
(6, 'Malang', 3);
```


- ms_Provinsi

```
INSERT INTO ms_Provinsi (Kode_provinsi, Nama) VALUES  
(1, 'Jawa Barat'),  
(2, 'Jawa Tengah'),  
(3, 'Jawa Timur');
```

- Hasil Insert
- tr_Pengiriman

MySQL Workbench interface showing the 'tr_pengiriman' table. The table structure is as follows:

Kode_Pengiriman	NamaPelanggan	AlamatAsal	AlamatTujuan	Kode_kecamatanAsal	Kode_kecamatanTujuan	Berat	TanggalPengiriman
1	Pelanggan 1	Jl. Merdeka No. 1	Jl. Sudirman No. 2	1	2	5.00	2023-07-30
2	Pelanggan 2	Jl. Pahlawan No. 3	Jl. Gatot Subroto No. 4	3	4	3.50	2023-07-30
3	Pelanggan 3	Jl. Diponegoro No. 5	Jl. Ahmad Yani No. 6	5	6	4.00	2023-07-30
4	Pelanggan 4	Jl. Gajah Mada No. 7	Jl. Hayam Wuruk No. 8	7	8	2.00	2023-07-30
5	Pelanggan 5	Jl. Thamrin No. 9	Jl. Kebon Sirih No. 10	9	1	6.00	2023-07-30
6	Pelanggan 6	Jl. Sudirman No. 11	Jl. MH Thamrin No. 12	2	3	3.00	2023-07-30
7	Pelanggan 7	Jl. Gatot Subroto No. 13	Jl. Pahlawan No. 14	4	5	4.50	2023-07-30
8	Pelanggan 8	Jl. Ahmad Yani No. 15	Jl. Diponegoro No. 16	6	7	5.50	2023-07-30
9	Pelanggan 9	Jl. Hayam Wuruk No. 17	Jl. Gajah Mada No. 18	8	9	2.50	2023-07-30
10	Pelanggan 10	Jl. Kebon Sirih No. 19	Jl. Thamrin No. 20	1	2	3.00	2023-07-30
11	Pelanggan 11	Jl. Merdeka No. 21	Jl. Sudirman No. 22	2	3	4.00	2023-07-30
12	Pelanggan 12	Jl. Pahlawan No. 23	Jl. Gatot Subroto No. 24	3	4	5.00	2023-07-30
13	Pelanggan 13	Jl. Dionegoro No. 25	Jl. Ahmad Yani No. 26	4	5	6.00	2023-07-30

- ms_StatusPembayaran

MySQL Workbench interface showing the 'ms_statuspembayaran' table. The table structure is as follows:

Kode_statuspembayaran	Nama
1	Lunas
2	Belum Lunas

- ms_Pembayaran

The screenshot shows the MySQL Workbench interface with the **ms_Pembayaran** table selected in the Navigator. The table structure is as follows:

Kode_pembayaran	Total	Kode_statuspembayaran	Kode_tipepembayaran
1	150000.00	1	1
2	200000.00	1	2
3	230000.00	2	3
4	300000.00	1	1
5	350000.00	1	2
6	400000.00	2	3
7	450000.00	1	1
8	500000.00	1	2
9	550000.00	2	3
10	600000.00	1	1
11	650000.00	1	2
12	700000.00	2	3
13	750000.00	1	1
14	800000.00	1	2

The Action Output pane shows the following query results:

#	Time	Action	Message	Duration / Fetch
2	07:31:51	SELECT * FROM phi_ekspedisi_ewan.ms_kota LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec
3	07:32:28	SELECT * FROM phi_ekspedisi_ewan.ms_kecamatan LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
4	07:33:08	SELECT * FROM phi_ekspedisi_ewan.ms_statuspengiriman LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
5	07:33:43	SELECT * FROM phi_ekspedisi_ewan.ms_kurir LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
6	07:34:23	SELECT * FROM phi_ekspedisi_ewan.ms_tipepembayaran LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
7	07:35:12	SELECT * FROM phi_ekspedisi_ewan.ms_pembayaran LIMIT 0, 1000	30 row(s) returned	0.000 sec / 0.000 sec

- ms_TipePembayaran

The screenshot shows the MySQL Workbench interface with the **ms_TipePembayaran** table selected in the Navigator. The table structure is as follows:

Kode_tipepembayaran	Nama
1	Transfer Bank
2	Kartu Kredit
3	Tunai

The Action Output pane shows the following query results:

#	Time	Action	Message	Duration / Fetch
1	07:31:05	SELECT * FROM phi_ekspedisi_ewan.ms_provinsi LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
2	07:31:51	SELECT * FROM phi_ekspedisi_ewan.ms_kota LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec
3	07:32:28	SELECT * FROM phi_ekspedisi_ewan.ms_kecamatan LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
4	07:33:08	SELECT * FROM phi_ekspedisi_ewan.ms_statuspengiriman LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
5	07:33:43	SELECT * FROM phi_ekspedisi_ewan.ms_kurir LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
6	07:34:23	SELECT * FROM phi_ekspedisi_ewan.ms_tipepembayaran LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

- ms_Kurir

The screenshot shows the MySQL Workbench interface. The left sidebar displays the database schema, including tables like `ms_kurir`, `ms_kota`, `ms_kecamatan`, `ms_statuspengiriman`, `ms_pembayaran`, `ms_provinsi`, `ms_tipepembayaran`, `ms_tipepengiriman`, `tr_pengiriman`, `tbl_pos`, `sampledatabase`, and `sys`. The main query editor shows a query: `SELECT * FROM phi_ekspedisi_evan.ms_kurir;`. The results are displayed in a table with columns: `Kode_kurir`, `Nama`, `TipeKendaraan`, and `NoHP`. The table contains 10 rows of data. The bottom panel shows the 'Action Output' for the query, indicating that 10 rows were returned.

Kode_kurir	Nama	TipeKendaraan	NoHP
1	Andi	Motor	081234567890
2	Budi	Mobil	081234567891
3	Cici	Motor	081234567892
4	Dedi	Mobil	081234567893
5	Eka	Motor	081234567894
6	Fani	Mobil	081234567895
7	Gani	Motor	081234567896
8	Hani	Mobil	081234567897
9	Ika	Motor	081234567898
10	Joni	Mobil	081234567899

- ms_StatusPengiriman

The screenshot shows the MySQL Workbench interface. The left sidebar displays the database schema, including tables like `ms_statuspengiriman`, `ms_kurir`, `ms_kota`, `ms_kecamatan`, `ms_pembayaran`, `ms_provinsi`, `ms_tipepembayaran`, `ms_tipepengiriman`, `tr_pengiriman`, `tbl_pos`, `sampledatabase`, and `sys`. The main query editor shows a query: `SELECT * FROM phi_ekspedisi_evan.ms_statuspengiriman;`. The results are displayed in a table with columns: `Kode_statuspengiriman` and `Nama`. The table contains 3 rows of data. The bottom panel shows the 'Action Output' for the query, indicating that 3 rows were returned.

Kode_statuspengiriman	Nama
1	Dalam Proses
2	Dalam Pengiriman
3	Terlirim

- ms_Kecamatan

The screenshot shows the MySQL Workbench interface with the `ms_ekspedisi_evan` database selected. The `ms_kecamatan` table is highlighted in the Navigator. The SQL editor contains the query: `SELECT * FROM phi_ekspedisi_evan.ms_kecamatan;`. The result grid shows 9 rows of data:

Kode_kecamatan	Nama	Kode_kota
1	Coblong	1
2	Subajadi	1
3	Harjanukti	2
4	Kejaksan	2
5	Semarang Tengah	3
6	Banjarsari	4
7	Jebres	4
8	Gubeng	5
9	Klojen	6

The Action Output pane shows the execution of three queries, with the third query (SELECT * FROM phi_ekspedisi_evan.ms_kecamatan LIMIT 0, 1000) returning 9 rows.

- ms_Kota

The screenshot shows the MySQL Workbench interface with the `ms_ekspedisi_evan` database selected. The `ms_kota` table is highlighted in the Navigator. The SQL editor contains the query: `SELECT * FROM phi_ekspedisi_evan.ms_kota;`. The result grid shows 6 rows of data:

Kode_kota	Nama	Kode_provinsi
1	Bandung	1
2	Cirebon	1
3	Semarang	2
4	Surakarta	2
5	Surabaya	3
6	Malang	3

The Action Output pane shows the execution of two queries, with the second query (SELECT * FROM phi_ekspedisi_evan.ms_kota LIMIT 0, 1000) returning 6 rows.

- ms_Provinsi

The screenshot shows the MySQL Workbench interface. The 'Query 1' window contains the SQL query: `SELECT * FROM phi_ekspedisi_evan.ms_provinsi;`. The 'Result Grid' displays the following data:

Kode_provinsi	Nama
1	Jawa Barat
2	Jawa Tengah
3	Jawa Timur

The 'Information' panel on the left shows the table structure for `ms_provinsi`:

Table: `ms_provinsi`
Columns:
`Kode_provinsi` int PK
`Nama` varchar(255)

The 'Action Output' panel at the bottom shows the execution details:

#	Time	Action	Message	Duration / Fetch
1	07:31:05	SELECT * FROM phi_ekspedisi_evan.ms_provinsi LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

2. Identifikasi table fact dan table dimension

- Table Fact

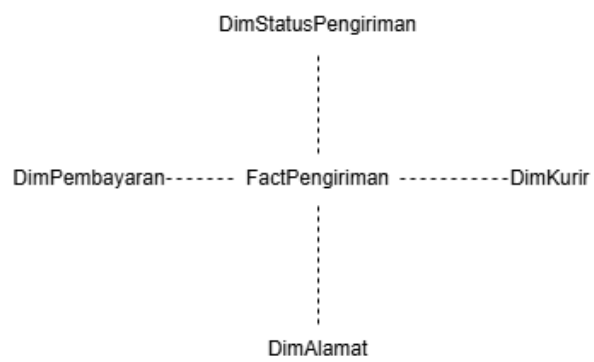
1. fact_pengiriman

- Kode_pengiriman
- NamaPelanggan
- AlamatAsal
- AlamatTujuan
- Kode_kecamatanAsal
- Kode_kecamatanTujuan
- Berat
- TanggalPengiriman
- TanggalSampaiPerkiraan
- TanggalSampaiAktual
- Kode_kurir
- NamaKurir
- Kode_statuspengiriman
- StatusPengiriman
- Kode_pembayaran
- TotalPembayaran

- Table Dimension

1. dim_alamat: sk_alamat, Kode_kecamatan, Nama_kecamatan, Kode_kota, Nama_kota, Kode_provinsi, Nama_provinsi
2. dim_pembayaran: sk_pembayaran, Kode_pembayaran, Kode_statuspembayaran, StatusPembayaran, Kode_tipepembayaran, TipePembayaran, Total
3. dim_kurir: Kode_kurir, NamaKurir, TipeKendaraan, NoHP
4. dim_statuspengiriman: Kode_pengiriman, StatusPengiriman

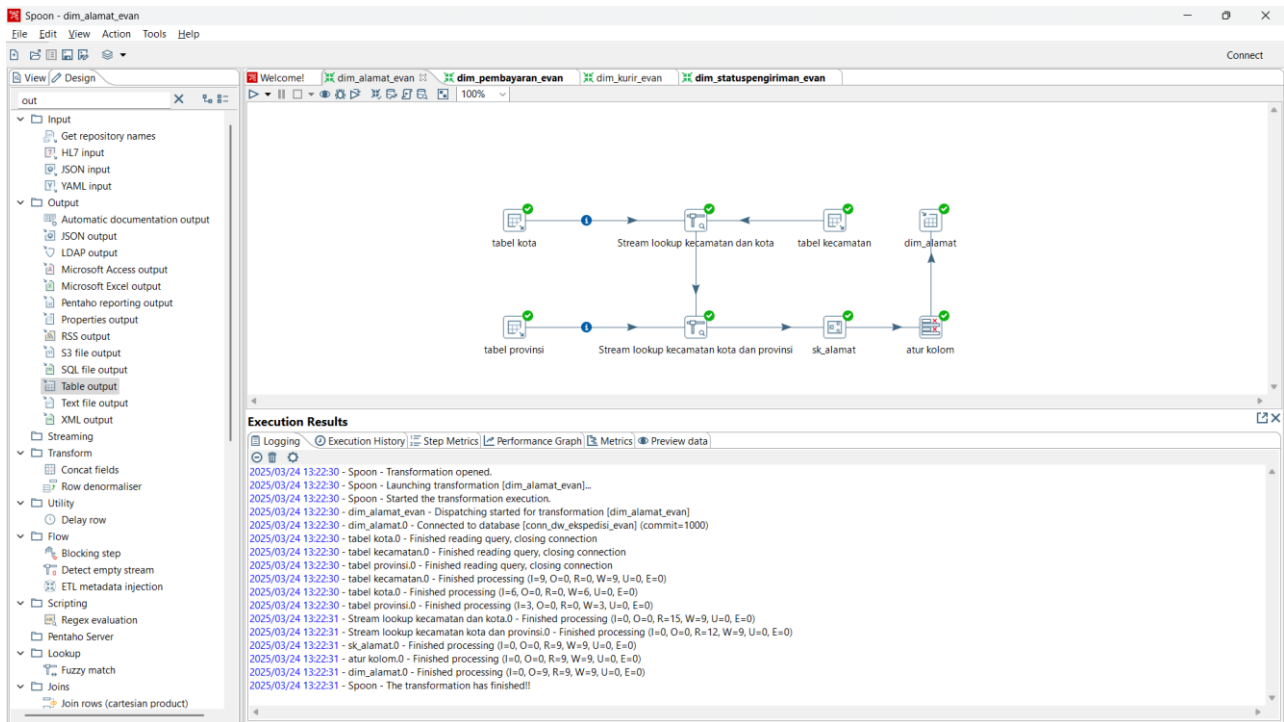
5. Gambar star schema database ekspedisi



6. Implementasi Star Schema di Pentaho Data Integration

- Table dim_alamat (dari penggabungan table kecamatan, table kota, dan table provinsi)

a. Struktur



b. Step Metrics

Execution Results

#	Stepname	Copynr	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	tabel kota	0	0	6	6	0	0	0	0	Finished	0.0s	857	-
2	tabel kecamatan	0	0	9	9	0	0	0	0	Finished	0.0s	3,000	-
3	Stream lookup kecamatan dan kota	0	15	9	0	0	0	0	0	Finished	0.2s	74	-
4	tabel provinsi	0	0	3	3	0	0	0	0	Finished	0.0s	428	-
5	Stream lookup kecamatan kota dan provinsi	0	12	9	0	0	0	0	0	Finished	0.2s	53	-
6	sk_alamat	0	9	9	0	0	0	0	0	Finished	0.3s	34	-
7	atur kolom	0	9	9	0	0	0	0	0	Finished	0.3s	31	-
8	dim_alamat	0	9	9	0	9	0	0	0	Finished	0.4s	23	-

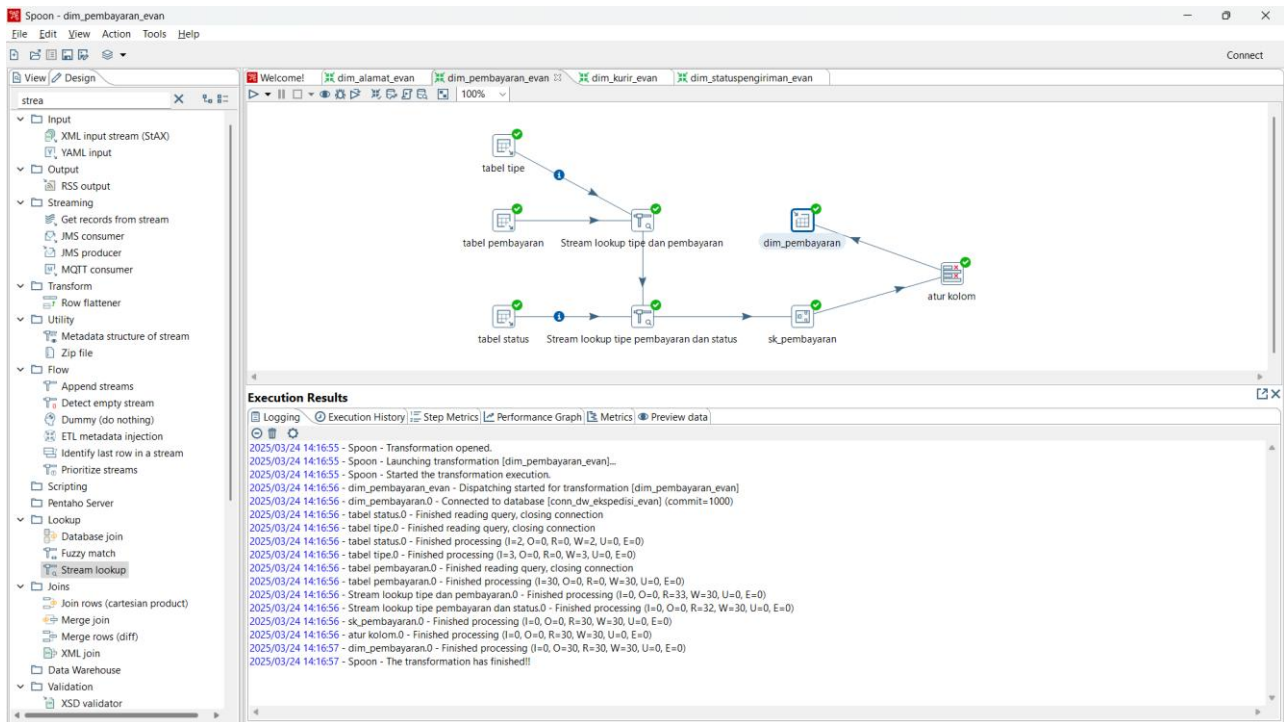
c. Preview Data

Execution Results

#	sk_alamat	Kode_provinsi	Nama_provinsi	Kode_kota	Nama_kota	Kode_kecamatan	Nama_kecamatan
1	1	1	Jawa Barat	1	Bandung	1	Coblong
2	2	1	Jawa Barat	1	Bandung	2	Sukajadi
3	3	1	Jawa Barat	2	Cirebon	3	Harjamukti
4	4	1	Jawa Barat	2	Cirebon	4	Kejaksan
5	5	2	Jawa Tengah	3	Semarang	5	Semarang Tengah
6	6	2	Jawa Tengah	4	Surakarta	6	Banjarsari
7	7	2	Jawa Tengah	4	Surakarta	7	Jebres
8	8	3	Jawa Timur	5	Surabaya	8	Gubeng
9	9	3	Jawa Timur	6	Malang	9	Klojen

- Table dim_pembayaran (dari gabungan table tipepembayaran, tablestatuspembayaran, dan table pembayaran)

a. Struktur



b. Step Metrics

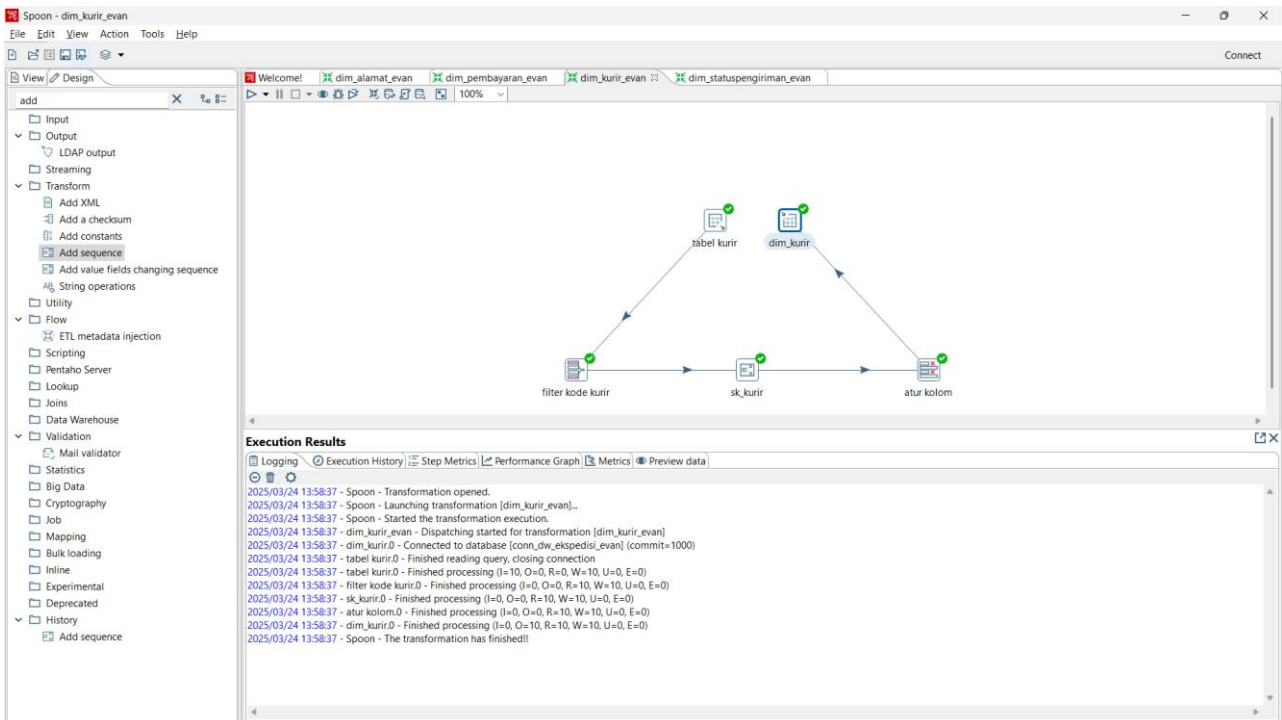
Execution Results													
Logging Execution History Step Metrics Performance Graph Metrics Preview data													
#	Stepname	Copyn	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	tabel tipe	0	0	3	3	0	0	0	0	Finished	0.0s	111	-
2	tabel pembayaran	0	0	30	30	0	0	0	0	Finished	0.0s	1,111	-
3	Stream lookup tipe dan pembayaran	0	33	30	0	0	0	0	0	Finished	0.2s	144	-
4	tabel status	0	0	2	2	0	0	0	0	Finished	0.0s	105	-
5	Stream lookup tipe pembayaran dan status	0	32	30	0	0	0	0	0	Finished	0.3s	120	-
6	sk_pembayaran	0	30	30	0	0	0	0	0	Finished	0.3s	103	-
7	atur kolom	0	30	30	0	0	0	0	0	Finished	0.3s	96	-
8	dim_pembayaran	0	30	30	0	30	0	0	0	Finished	0.4s	70	-

c. Preview Data

Execution Results							
Logging Execution History Step Metrics Performance Graph Metrics Preview data First rows Last rows Off							
#	sk_pembayaran	Kode_tipepembayaran	TipePembayaran	Kode_statuspembayaran	StatusPembayaran	Kode_pembayaran	Total
1	1	1	Transfer Bank	1	Lunas	1	150000.0
2	2	2	Kartu Kredit	1	Lunas	2	200000.0
3	3	3	Tunai	2	Belum Lunas	3	250000.0
4	4	1	Transfer Bank	1	Lunas	4	300000.0
5	5	2	Kartu Kredit	1	Lunas	5	350000.0
6	6	3	Tunai	2	Belum Lunas	6	400000.0
7	7	1	Transfer Bank	1	Lunas	7	450000.0
8	8	2	Kartu Kredit	1	Lunas	8	500000.0
9	9	3	Tunai	2	Belum Lunas	9	550000.0
10	10	1	Transfer Bank	1	Lunas	10	600000.0
11	11	2	Kartu Kredit	1	Lunas	11	650000.0
12	12	3	Tunai	2	Belum Lunas	12	700000.0
13	13	1	Transfer Bank	1	Lunas	13	750000.0
14	14	2	Kartu Kredit	1	Lunas	14	800000.0
15	15	3	Tunai	2	Belum Lunas	15	850000.0
16	16	1	Transfer Bank	1	Lunas	16	900000.0
17	17	2	Kartu Kredit	1	Lunas	17	950000.0
18	18	3	Tunai	2	Belum Lunas	18	1000000.0
19	19	1	Transfer Bank	1	Lunas	19	1050000.0
20	20	2	Kartu Kredit	1	Lunas	20	1100000.0
21	21	3	Tunai	2	Belum Lunas	21	1150000.0
22	22	1	Transfer Bank	1	Lunas	22	1200000.0
23	23	2	Kartu Kredit	1	Lunas	23	1250000.0
24	24	3	Tunai	2	Belum Lunas	24	1300000.0
25	25	1	Transfer Bank	1	Lunas	25	1350000.0
26	26	2	Kartu Kredit	1	Lunas	26	1400000.0
27	27	3	Tunai	2	Belum Lunas	27	1450000.0
28	28	1	Transfer Bank	1	Lunas	28	1500000.0
29	29	2	Kartu Kredit	1	Lunas	29	1550000.0
30	30	3	Tunai	2	Belum Lunas	30	1600000.0

- Table dim_kurir

a. Struktur



b. Step Metrics

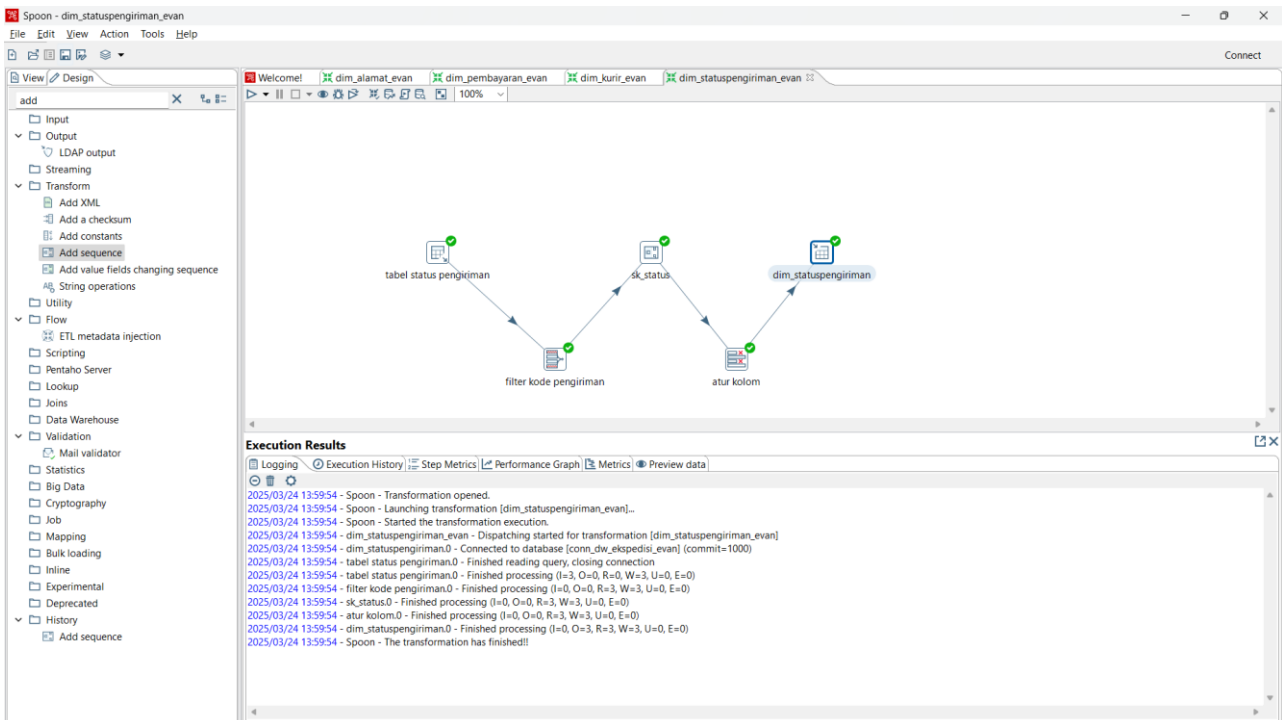
Execution Results													
Logging Execution History Step Metrics Performance Graph Metrics Preview data													
#	Stepname	Copynr	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	tabel kurir	0	0	10	10	0	0	0	0	Finished	0.0s	2,000	-
2	filter kode kurir	0	10	10	0	0	0	0	0	Finished	0.0s	385	-
3	sk_kurir	0	10	10	0	0	0	0	0	Finished	0.0s	227	-
4	atur kolom	0	10	10	0	0	0	0	0	Finished	0.1s	164	-
5	dim_kurir	0	10	10	0	10	0	0	0	Finished	0.2s	61	-

c. Preview Data

Execution Results					
Logging Execution History Step Metrics Performance Graph Metrics Preview data					
First rows Last rows Off					
#	sk_kurir	Kode_kurir	NamaKurir	TipeKendaraan	NoHP
1	1	1	Andi	Motor	081234567890
2	2	2	Budi	Mobil	081234567891
3	3	3	Cici	Motor	081234567892
4	4	4	Dedi	Mobil	081234567893
5	5	5	Eka	Motor	081234567894
6	6	6	Fani	Mobil	081234567895
7	7	7	Gani	Motor	081234567896
8	8	8	Hani	Mobil	081234567897
9	9	9	Ika	Motor	081234567898
10	10	10	Joni	Mobil	081234567899

- Table dim_statuspengiriman

a. Struktur



b. Step Metrics

Execution Results													
Logging Execution History Step Metrics Performance Graph Metrics Preview data													
#	Stepname	Copynr	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	tabel status pengiriman	0	0	3	3	0	0	0	0	Finished	0.0s	375	-
2	filter kode pengiriman	0	3	3	0	0	0	0	0	Finished	0.0s	75	-
3	sk_status	0	3	3	0	0	0	0	0	Finished	0.1s	51	-
4	atur kolom	0	3	3	0	0	0	0	0	Finished	0.1s	33	-
5	dim_statuspengiriman	0	3	3	0	3	0	0	0	Finished	0.1s	26	-

c. Preview Data

Execution Results			
Logging Execution History Step Metrics Performance Graph Metrics Preview data			
First rows Last rows Off			
#	sk_status	Kode_statuspengiriman	StatusPengiriman
1	1	1	Dalam Proses
2	2	2	Dalam Pengiriman
3	3	3	Terkirim

- Table fact_pengiriman

a. Struktur



b. Step Metrics

#	Stepname	Copynr	Read	Written	Input	Output	Updated	Rejected	Errors	Active	Time	Speed (r/s)	input/output
1	dim_kurir	0	0	10	10	0	0	0	0	Finished	0.0s	1,250	-
2	dim_statuspengiriman	0	0	3	3	0	0	0	0	Finished	0.0s	375	-
3	dim_alamat	0	0	9	9	0	0	0	0	Finished	0.0s	1,125	-
4	tr_pengiriman	0	0	30	30	0	0	0	0	Finished	0.0s	3,000	-
5	sk_alamat	0	39	30	0	0	0	0	0	Finished	0.2s	191	-
6	sk_kurir	0	40	30	0	0	0	0	0	Finished	0.2s	182	-
7	dim_pembayaran	0	0	30	30	0	0	0	0	Finished	0.0s	3,000	-
8	sk_pembayaran	0	60	30	0	0	0	0	0	Finished	0.2s	247	-
9	sk_status	0	33	30	0	0	0	0	0	Finished	0.3s	123	-
10	atur kolom	0	30	30	0	0	0	0	0	Finished	0.3s	106	-
11	fact_pengiriman	0	30	30	0	30	0	0	0	Finished	0.4s	79	-

c. Preview Data

Execution Results									
Logging Execution History Step Metrics Performance Graph Metrics Preview data ☒ First rows ☐ Last rows ☐ Off									
#	sk_alamat	sk_kurir	sk_pembayaran	sk_status	Kode_Pengiriman	NamaPelanggan	TanggalPengiriman	TanggalSampaiAktual	Total
1	<null>	1	1	1	1	Pelanggan 1	2023/10/01 00:00:00.000	2023/10/03 00:00:00.000	150000.0
2	<null>	2	2	3	2	Pelanggan 2	2023/10/02 00:00:00.000	2023/10/04 00:00:00.000	200000.0
3	<null>	3	3	2	3	Pelanggan 3	2023/10/03 00:00:00.000	2023/10/05 00:00:00.000	250000.0
4	<null>	4	4	1	4	Pelanggan 4	2023/10/04 00:00:00.000	2023/10/06 00:00:00.000	300000.0
5	<null>	5	5	3	5	Pelanggan 5	2023/10/05 00:00:00.000	2023/10/07 00:00:00.000	350000.0
6	<null>	6	6	2	6	Pelanggan 6	2023/10/06 00:00:00.000	2023/10/08 00:00:00.000	400000.0
7	<null>	7	7	1	7	Pelanggan 7	2023/10/07 00:00:00.000	2023/10/09 00:00:00.000	450000.0
8	<null>	8	8	3	8	Pelanggan 8	2023/10/08 00:00:00.000	2023/10/10 00:00:00.000	500000.0
9	<null>	9	9	2	9	Pelanggan 9	2023/10/09 00:00:00.000	2023/10/11 00:00:00.000	550000.0
10	<null>	10	10	1	10	Pelanggan 10	2023/10/10 00:00:00.000	2023/10/12 00:00:00.000	600000.0
11	<null>	1	11	3	11	Pelanggan 11	2023/10/11 00:00:00.000	2023/10/13 00:00:00.000	650000.0
12	<null>	2	12	2	12	Pelanggan 12	2023/10/12 00:00:00.000	2023/10/14 00:00:00.000	700000.0
13	<null>	3	13	1	13	Pelanggan 13	2023/10/13 00:00:00.000	2023/10/15 00:00:00.000	750000.0
14	<null>	4	14	3	14	Pelanggan 14	2023/10/14 00:00:00.000	2023/10/16 00:00:00.000	800000.0
15	<null>	5	15	2	15	Pelanggan 15	2023/10/15 00:00:00.000	2023/10/17 00:00:00.000	850000.0
16	<null>	6	16	1	16	Pelanggan 16	2023/10/16 00:00:00.000	2023/10/18 00:00:00.000	900000.0
17	<null>	7	17	3	17	Pelanggan 17	2023/10/17 00:00:00.000	2023/10/19 00:00:00.000	950000.0
18	<null>	8	18	2	18	Pelanggan 18	2023/10/18 00:00:00.000	2023/10/20 00:00:00.000	1000000.0
19	<null>	9	19	1	19	Pelanggan 19	2023/10/19 00:00:00.000	2023/10/21 00:00:00.000	1050000.0
20	<null>	10	20	3	20	Pelanggan 20	2023/10/20 00:00:00.000	2023/10/22 00:00:00.000	1100000.0
21	<null>	1	21	2	21	Pelanggan 21	2023/10/21 00:00:00.000	2023/10/23 00:00:00.000	1150000.0
22	<null>	2	22	1	22	Pelanggan 22	2023/10/22 00:00:00.000	2023/10/24 00:00:00.000	1200000.0
23	<null>	3	23	3	23	Pelanggan 23	2023/10/23 00:00:00.000	2023/10/25 00:00:00.000	1250000.0
24	<null>	4	24	2	24	Pelanggan 24	2023/10/24 00:00:00.000	2023/10/26 00:00:00.000	1300000.0
25	<null>	5	25	1	25	Pelanggan 25	2023/10/25 00:00:00.000	2023/10/27 00:00:00.000	1350000.0
26	<null>	6	26	3	26	Pelanggan 26	2023/10/26 00:00:00.000	2023/10/28 00:00:00.000	1400000.0
27	<null>	2	27	3	27	Pelanggan 27	2023/10/27 00:00:00.000	2023/10/29 00:00:00.000	1450000.0
28	<null>	8	28	1	28	Pelanggan 28	2023/10/28 00:00:00.000	2023/10/30 00:00:00.000	1500000.0
29	<null>	9	29	3	29	Pelanggan 29	2023/10/29 00:00:00.000	2023/10/31 00:00:00.000	1550000.0
30	<null>	10	30	2	30	Pelanggan 30	2023/10/30 00:00:00.000	2023/11/01 00:00:00.000	1600000.0

- Hasil dari pembuatan star schema di pentaho
- Table dim_alamat

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'dw_ekspedisi_ewan' selected. The 'dim_alamat' table is highlighted. The main window shows the table's structure and a query result.

Table: dim_alamat

Columns:

- sk_alamat: int
- Kode_provinsi: int
- Nama_provinsi: varchar(2)
- Kode_kota: int
- Nama_kota: varchar(2)
- Kode_kecamatan: int
- Nama_kecamatan: varchar(2)

Query Result:

sk_alamat	Kode_provinsi	Nama_provinsi	Kode_kota	Nama_kota	Kode_kecamatan	Nama_kecamatan
1	1	Java Barat	1	Bandung	1	Ciblong
2	1	Java Barat	1	Bandung	2	Sukajadi
3	1	Java Barat	2	Cirebon	3	Harjamukti
4	1	Java Barat	2	Cirebon	4	Kejaksan
5	2	Java Tengah	3	Semarang	5	Semarang Tengah
6	2	Java Tengah	4	Surakarta	6	Barjarsari
7	2	Java Tengah	4	Surakarta	7	Jedines
8	3	Java Timur	5	Surabaya	8	Gubeng
9	3	Java Timur	6	Malang	9	Klojen

Action Output:

#	Time	Action	Message	Duration / Fetch
5	13:36:46	SELECT * FROM phi_ekspedi_ewan.ms_kurir LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
6	13:36:52	SELECT * FROM phi_ekspedi_ewan.ms_tipepembayaran LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
7	13:37:03	SELECT * FROM phi_ekspedi_ewan.ms_statuspembayaran LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
8	13:37:06	SELECT * FROM phi_ekspedi_ewan.ms_pembayaran LIMIT 0, 1000	30 row(s) returned	0.000 sec / 0.000 sec
9	13:37:14	SELECT * FROM phi_ekspedi_ewan.tr_pengiriman LIMIT 0, 1000	30 row(s) returned	0.000 sec / 0.000 sec
10	13:37:20	SELECT * FROM dw_ekspedi_ewan.dim_alamat LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec

- Table dim_pembayaran

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'dw_ekspedisi_ewan' selected. The 'dim_pembayaran' table is highlighted. The main window shows the table's structure and a query result.

Table: dim_pembayaran

Columns:

- sk_pembayaran: int
- Kode_tipepembayaran: int
- TipePembayaran: varchar(2)
- Kode_statuspembayaran: int
- StatusPembayaran: varchar(2)
- Kode_pembayaran: int
- Total: int

Query Result:

sk_pembayaran	Kode_tipepembayaran	TipePembayaran	Kode_statuspembayaran	StatusPembayaran	Kode_pembayaran	Total
1	1	Transfer Bank	1	Lunas	1	150000
2	2	Kartu Kredit	1	Lunas	2	200000
3	3	Tunai	2	Belum Lunas	3	250000
4	1	Transfer Bank	1	Lunas	4	300000
5	2	Kartu Kredit	1	Lunas	5	350000
6	3	Tunai	2	Belum Lunas	6	400000
7	1	Transfer Bank	1	Lunas	7	450000
8	2	Kartu Kredit	1	Lunas	8	500000
9	3	Tunai	2	Belum Lunas	9	550000
10	1	Transfer Bank	1	Lunas	10	600000
11	2	Kartu Kredit	1	Lunas	11	650000
12	3	Tunai	2	Belum Lunas	12	700000
13	1	Transfer Bank	1	Lunas	13	750000
14	2	Kartu Kredit	1	Lunas	14	800000
15	3	Tunai	2	Belum Lunas	15	850000
16	1	Transfer Bank	1	Lunas	16	900000
17	2	Kartu Kredit	1	Lunas	17	950000
18	3	Tunai	2	Belum Lunas	18	1000000

Action Output:

#	Time	Action	Message	Duration / Fetch
1	14:18:59	SELECT * FROM dw_ekspedi_ewan.dim_pembayaran LIMIT 0, 1000	30 row(s) returned	0.000 sec / 0.000 sec

- Table dim_kurir

MySQL Workbench interface showing the **dim_kurir** table structure and query results.

Table: dim_kurir

Columns:

- sk_kurir: int
- Kode_kurir: int
- NamaKurir: varchar(255)
- TipeKendaraan: varchar(255)
- NoHP: varchar(15)

Query 1: `SELECT * FROM dw_ekspedisi_ewan.dim_kurir;`

Result Grid:

sk_kurir	Kode_kurir	NamaKurir	TipeKendaraan	NoHP
1	1	Andi	Motor	081234567890
2	2	Budi	Mobil	081234567891
3	3	Cici	Motor	081234567892
4	4	Dedi	Mobil	081234567893
5	5	Eka	Motor	081234567894
6	6	Fani	Mobil	081234567895
7	7	Gani	Motor	081234567896
8	8	Hani	Mobil	081234567897
9	9	Ika	Motor	081234567898
10	10	Joni	Mobil	081234567899

Action Output:

#	Time	Action	Message	Duration / Fetch
1	14:01:51	SELECT * FROM dw_ekspedisi_ewan.dim_kurir LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

- Table dim_statuspembayaran

MySQL Workbench interface showing the **dim_statuspembayaran** table structure and query results.

Table: dim_statuspembayaran

Columns:

- sk_status: int
- Kode_statuspembayaran: int
- StatusPembayaran: varchar

Query 1: `SELECT * FROM dw_ekspedisi_ewan.dim_statuspembayaran;`

Result Grid:

sk_status	Kode_statuspembayaran	StatusPembayaran
1	1	Dalam Proses
2	2	Dalam Pengiriman
3	3	Terkin

Action Output:

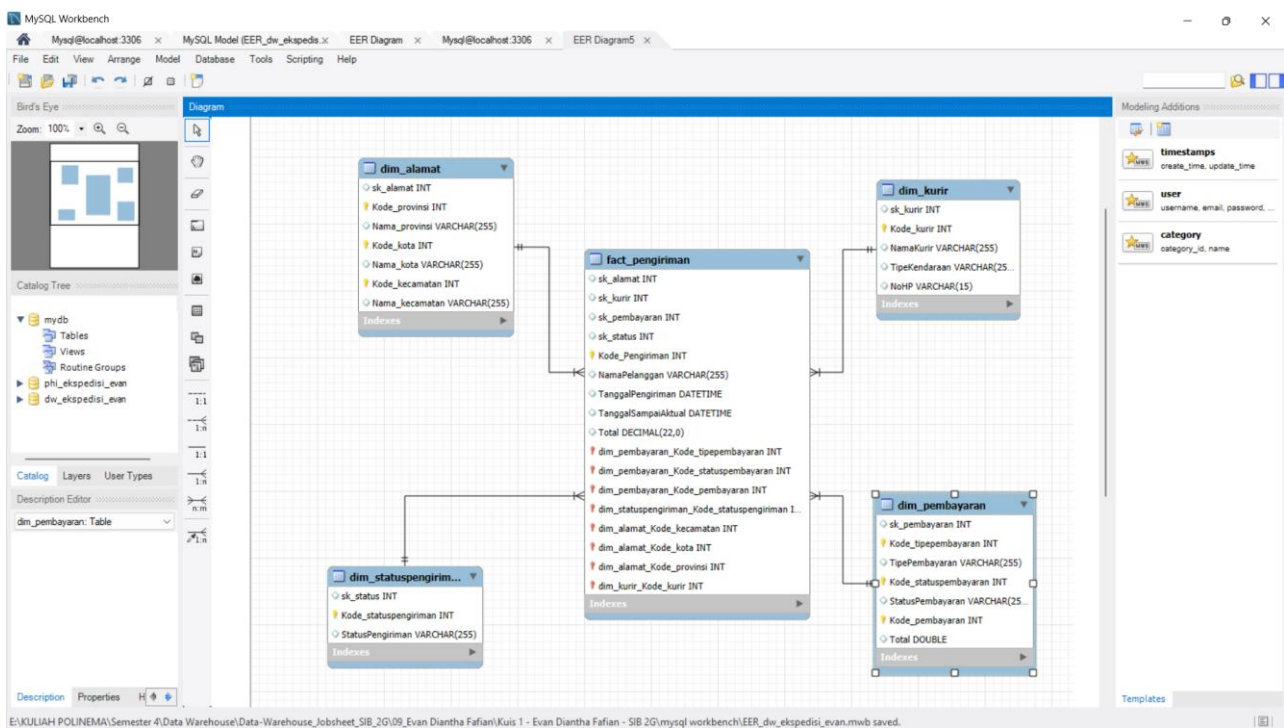
#	Time	Action	Message	Duration / Fetch
1	14:01:51	SELECT * FROM dw_ekspedisi_ewan.dim_kurir LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
2	14:02:33	SELECT * FROM dw_ekspedisi_ewan.dim_statuspembayaran LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

- Table fact_pengiriman

Query 1: `SELECT * FROM dw_ekspedisi_ewan.fact_pengiriman;`

sk_alamat	sk_kurir	sk_pembayaran	sk_status	Kode_Pengiriman	NamaPelanggan	TanggalPengiriman	TanggalSampaiAktual	Total
1	1	1	1	1	Pelanggan 1	2023-10-01 00:00:00	2023-10-03 00:00:00	150000
2	2	2	3	2	Pelanggan 2	2023-10-02 00:00:00	2023-10-04 00:00:00	200000
3	3	3	2	3	Pelanggan 3	2023-10-03 00:00:00	2023-10-05 00:00:00	250000
4	4	4	1	4	Pelanggan 4	2023-10-04 00:00:00	2023-10-06 00:00:00	300000
5	5	5	3	5	Pelanggan 5	2023-10-05 00:00:00	2023-10-07 00:00:00	350000
6	6	6	2	6	Pelanggan 6	2023-10-06 00:00:00	2023-10-08 00:00:00	400000
7	7	7	1	7	Pelanggan 7	2023-10-07 00:00:00	2023-10-09 00:00:00	450000
8	8	8	3	8	Pelanggan 8	2023-10-08 00:00:00	2023-10-10 00:00:00	500000
9	9	9	2	9	Pelanggan 9	2023-10-09 00:00:00	2023-10-11 00:00:00	550000
10	10	10	1	10	Pelanggan 10	2023-10-10 00:00:00	2023-10-12 00:00:00	600000
11	11	11	3	11	Pelanggan 11	2023-10-11 00:00:00	2023-10-13 00:00:00	650000
12	12	12	2	12	Pelanggan 12	2023-10-12 00:00:00	2023-10-14 00:00:00	700000
13	13	13	1	13	Pelanggan 13	2023-10-13 00:00:00	2023-10-15 00:00:00	750000
14	14	14	3	14	Pelanggan 14	2023-10-14 00:00:00	2023-10-16 00:00:00	800000
15	15	15	2	15	Pelanggan 15	2023-10-15 00:00:00	2023-10-17 00:00:00	850000
16	16	16	1	16	Pelanggan 16	2023-10-16 00:00:00	2023-10-18 00:00:00	900000
17	17	17	3	17	Pelanggan 17	2023-10-17 00:00:00	2023-10-19 00:00:00	950000
18	18	18	2	18	Pelanggan 18	2023-10-18 00:00:00	2023-10-20 00:00:00	1000000

- EER star schema ekspedisi



dw_ekpedisi_ewan